

# Signals, Systems and Transforms

By Dr Nafaa NACEREDDINE

# Chapter 5

## z-Transform

### Chapter Intended Learning Outcomes:

- (i) Understanding the relationship between z-transform and the Fourier transform for discrete-time signals
- (ii) Understanding the characteristics and properties of z-transform
- (iii) Ability to compute z-transform and inverse z-transform
- (iv) Ability to apply z-transform for analyzing linear time-invariant (LTI) systems

### References

- J.G.Proakis and D.G.Manolakis, *Digital Signal Processing: Principles, Algorithms, and Applications*, Prentice Hall 2007.
- A.V.Oppenheim and R.W.Schafer, *Discrete-Time Signal Processing*, Prentice Hall, 1998.
- Hing Cheung So, *Lecture Notes on Digital Signal Processing*, Lectures at DEE of City University of Hong Kong, Feb. 2016.
- Mark Fowler, *Digital Signal Processing*, Lectures at Binghamton University, NY, 2015

## Definition

The  $z$  transform of  $x[n]$ , denoted by  $X(z)$ , is defined as:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad (5.1)$$

where  $z$  is a **continuous complex** variable.

**Is  $X(z)$  real-valued or complex-valued?**

## Relationship with Fourier Transform

As seen in Chapter 4, we construct the continuous-time sampled signal  $x_s(t)$  with a sampling interval of  $T$  from  $x[n]$ :

$$x_s(t) = \sum_{n=-\infty}^{\infty} x[n]\delta(t - nT) \quad (5.2)$$

Taking Fourier transform of  $x_s(t)$  with using properties of  $\delta(t)$ :

$$\begin{aligned} X_s(j\Omega) &= \int_{-\infty}^{\infty} x_s(t) e^{-j\Omega t} dt = \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} x[n] \delta(t - nT) e^{-j\Omega t} dt \\ &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega nT} \end{aligned} \quad (5.3)$$

Defining  $\omega = \Omega T$  as the **discrete-time frequency** parameter and writing  $X_s(j\Omega)$  as  $X(e^{j\omega})$ , (5.3) becomes

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \quad (5.4)$$

which is known as **discrete-time Fourier transform (DTFT)** or **Fourier transform of discrete-time signals**

$X(e^{j\omega})$  is periodic with period  $2\pi$ :

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x[n]e^{-j(\omega+2k\pi)n} = X(e^{j(\omega+2k\pi)}) \quad (5.5)$$

where  $k$  is any integer. Since  $z$  is a continuous complex variable, we can write

$$z = re^{j\omega} \quad (5.6)$$

where  $r = |z| > 0$  is magnitude and  $\omega = \angle(z)$  is angle of  $z$ . Employing (5.6), the  $z$  transform is:

$$X(z)|_{z=re^{j\omega}} = X(re^{j\omega}) = \sum_{n=-\infty}^{\infty} (x[n]r^{-n}) e^{-j\omega n} \quad (5.7)$$

which is equal to the DTFT of  $x[n]r^{-n}$ . When  $r = 1$  or  $z = e^{j\omega}$ , (5.7) and (5.4) are identical:

$$X(z)|_{z=e^{j\omega}} = X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad (5.8)$$

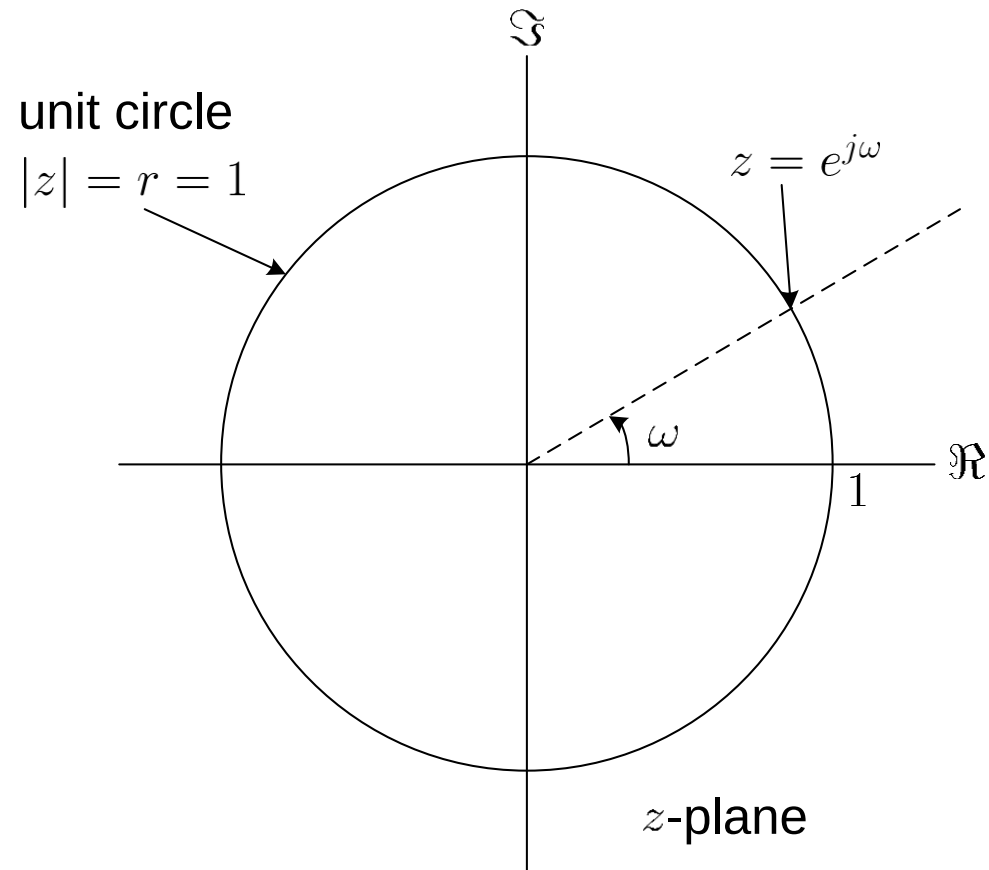


Fig.5.1: Relationship between  $X(z)$  and  $X(e^{j\omega})$  on the  $z$ -plane

## Region of Convergence (ROC)

ROC indicates when  $z$  transform of a sequence converges

Generally there exists some  $z$  such that

$$|X(z)| = \left| \sum_{n=-\infty}^{\infty} x[n]z^{-n} \right| \rightarrow \infty \quad (5.9)$$

where the  $z$  transform does not converge

The set of values of  $z$  for which  $X(z)$  converges or

$$|X(z)| = \left| \sum_{n=-\infty}^{\infty} x[n]z^{-n} \right| \leq \sum_{n=-\infty}^{\infty} |x[n]z^{-n}| < \infty \quad (5.10)$$

is called the ROC, which must be specified along with  $X(z)$  in order for the  $z$  transform to be complete

Assuming that  $x[n]$  is of infinite length, we decompose  $X(z)$ :

$$X(z) = X_-(z) + X_+(z) \quad (5.11)$$

where

$$X_-(z) = \sum_{n=-\infty}^{-1} x[n]z^{-n} = \sum_{m=1}^{\infty} x[-m]z^m \quad (5.12)$$

and

$$X_+(z) = \sum_{n=0}^{\infty} x[n]z^{-n} \quad (5.13)$$

Let  $f_n(z) = x[n]z^{-n}$ ,  $X_+(z)$  is expanded as:

$$\begin{aligned} X_+(z) &= x[0]z^{-0} + x[1]z^{-1} + \cdots + x[n]z^{-n} + \cdots \\ &= f_0(z) + f_1(z) + \cdots + f_n(z) + \cdots \end{aligned} \quad (5.14)$$



According to the ratio test, convergence of  $X_+(z)$  requires

$$\lim_{n \rightarrow \infty} \left| \frac{f_{n+1}(z)}{f_n(z)} \right| < 1 \quad (5.15)$$

**Let**  $\lim_{n \rightarrow \infty} |x[n+1]/x[n]| = R_+ > 0$ .  $X_+(z)$  **converges if**

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{x[n+1]z^{-n-1}}{x[n]z^{-n}} \right| &= \lim_{n \rightarrow \infty} \left| \frac{x[n+1]}{x[n]} \right| |z^{-1}| < 1 \\ \Rightarrow |z| &> \lim_{n \rightarrow \infty} \left| \frac{x[n+1]}{x[n]} \right| = R_+ \end{aligned} \quad (5.16)$$

That is, the ROC for  $X_+(z)$  is  $|z| > R_+$ .

Let  $\lim_{m \rightarrow \infty} |x[-m]/x[-m-1]| = R_- > 0$ .  $X_-(z)$  converges if

$$\begin{aligned} \lim_{m \rightarrow \infty} \left| \frac{x[-m-1]z^{m+1}}{x[-m]z^m} \right| &= \lim_{m \rightarrow \infty} \left| \frac{x[-m-1]}{x[-m]} \right| |z| < 1 \\ \Rightarrow |z| &< \lim_{m \rightarrow \infty} \left| \frac{x[-m]}{x[-m-1]} \right| = R_- \end{aligned} \quad (5.17)$$

As a result, the ROC for  $X_-(z)$  is  $|z| < R_-$

Combining the results, the ROC for  $X(z)$  is  $R_+ < |z| < R_-$ :

- ROC is a **ring** when  $R_+ < R_-$
- **No ROC** if  $R_- < R_+$  and  $X(z)$  **does not exist**

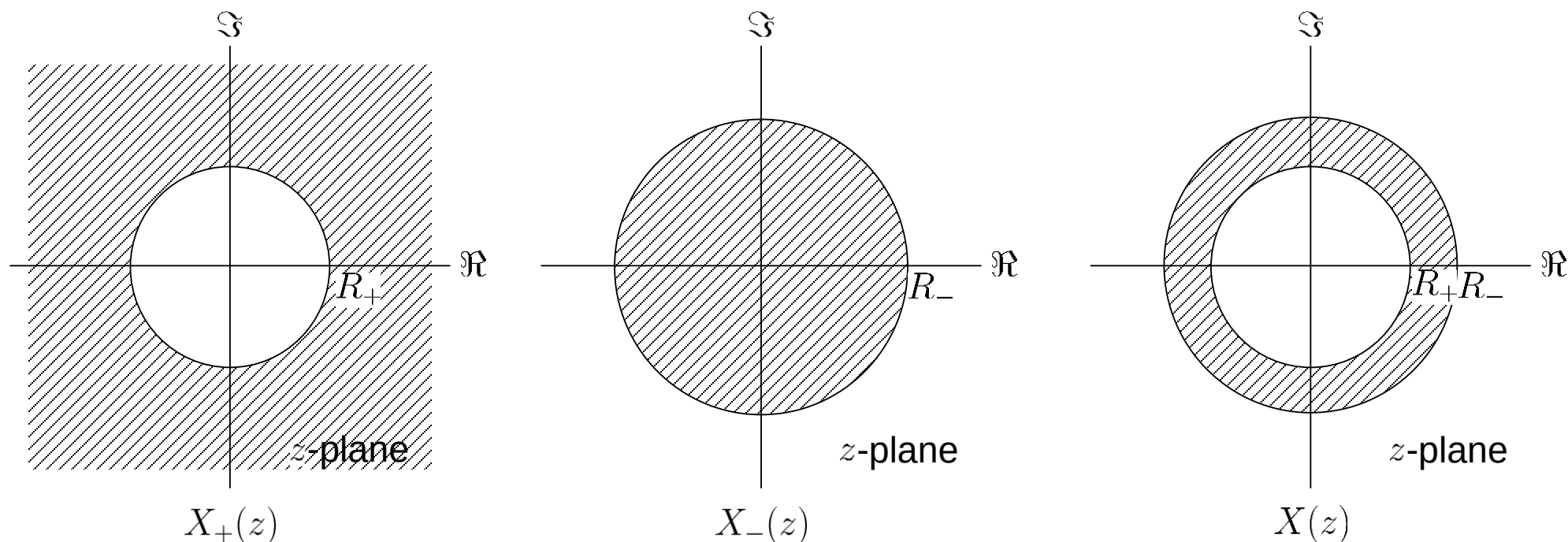


Fig.5.2: ROCs for  $X_+(z)$ ,  $X_-(z)$  and  $X(z)$

## Poles and Zeros

Values of  $z$  for which  $X(z) = 0$  are the **zeros** of  $X(z)$

Values of  $z$  for which  $X(z) = \infty$  are the **poles** of  $X(z)$

In many real-world applications,  $X(z)$  is represented as a rational function:

$$X(z) = \frac{P(z)}{Q(z)} = \frac{\sum_{k=0}^M b_k z^k}{\sum_{k=0}^N a_k z^k} \quad (5.18)$$

Factorizing  $P(z)$  and  $Q(z)$ , (5.18) can be written as

$$X(z) = \frac{b_0(z - d_1)(z - d_2) \cdots (z - d_M)}{a_0(z - c_1)(z - c_2) \cdots (z - c_N)} \quad (5.19)$$

**How many poles and zeros in (5.18)? What are they?**

### Example 5.1

Determine the  $z$  transform of  $x[n] = a^n u[n]$  where  $u[n]$  is the unit step function. Then determine the condition when the DTFT of  $x[n]$  exists.

Using (5.1) and (3.3), we have

$$X(z) = \sum_{n=-\infty}^{\infty} a^n u[n] z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n$$

According to (5.10),  $X(z)$  converges if

$$\sum_{n=0}^{\infty} |az^{-1}|^n < \infty$$

Applying the ratio test, the convergence condition is

$$|az^{-1}| < 1 \Leftrightarrow |z| > |a|$$

Note that we cannot write  $|z| > a$  because  $a$  may be complex

For  $|z| > |a|$ ,  $X(z)$  is computed as

$$X(z) = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1 - (az^{-1})^{\infty}}{1 - az^{-1}} = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}$$

Together with the ROC, the  $z$  transform of  $x[n] = a^n u[n]$  is:

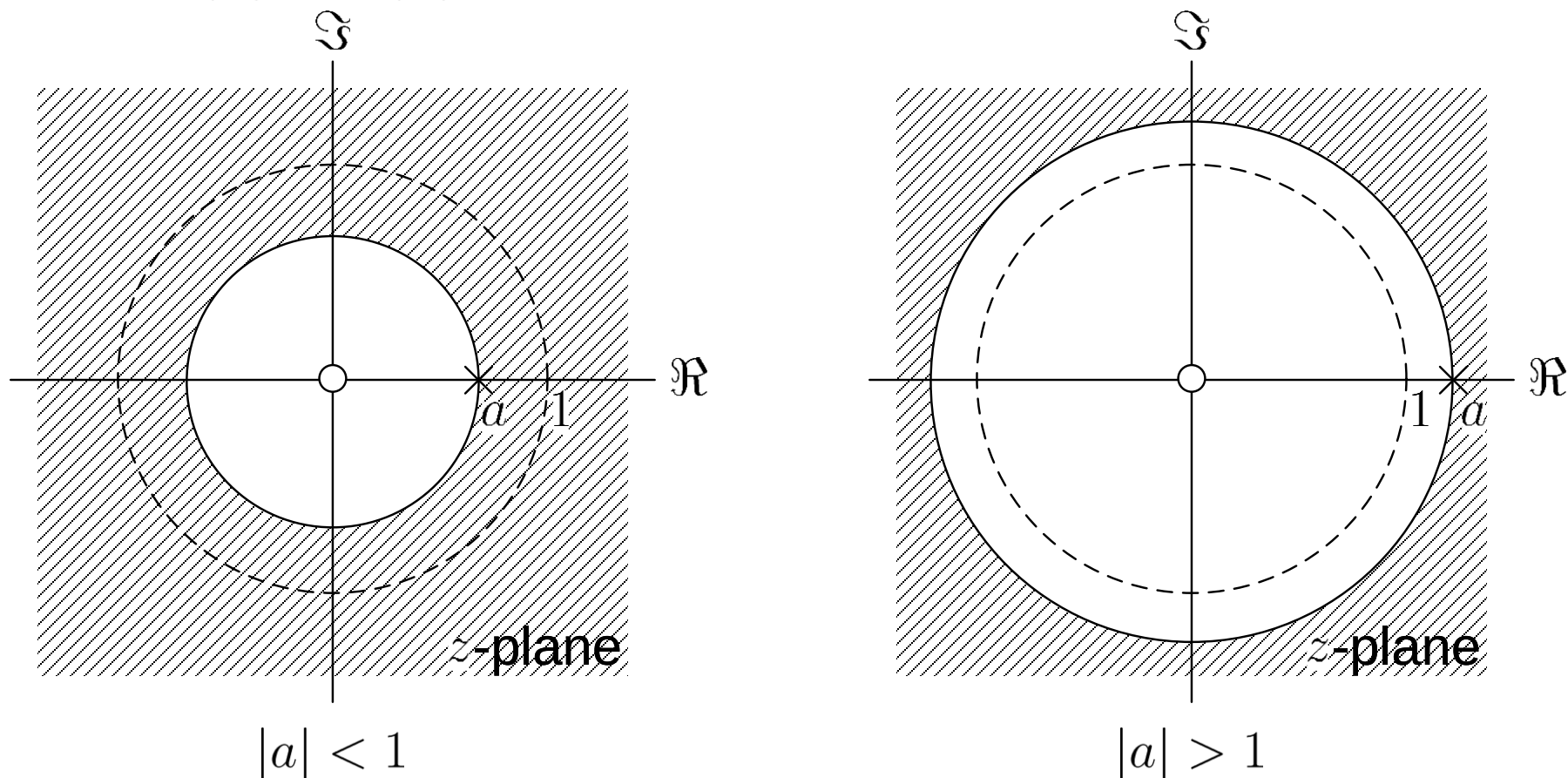
$$X(z) = \frac{z}{z - a}, \quad |z| > |a|$$

It is clear that  $X(z)$  has a zero at  $z = 0$  and a pole at  $z = a$ . Using (5.8), we substitute  $z = e^{j\omega}$  to obtain

$$X(e^{j\omega}) = \frac{e^{j\omega}}{e^{j\omega} - a}, \quad |e^{j\omega}| = 1 > |a|$$

As a result, the existence condition for DTFT of  $x[n]$  is  $|a| < 1$ .

Otherwise, its DTFT does not exist. In general, the DTFT  $X(e^{j\omega})$  exists if its **ROC includes the unit circle**. If  $|z| > |a|$  includes  $|z| = 1$ ,  $|a| < 1$  is required.



**Fig.5.3: ROCs for  $|a| < 1$  and  $|a| > 1$  when  $x[n] = a^n u[n]$**

### Example 5.2

Determine the  $z$  transform of  $x[n] = -a^n u[-n - 1]$ . Then determine the condition when the DTFT of  $x[n]$  exists.

Using (5.1) and (3.3), we have

$$X(z) = \sum_{n=-\infty}^{-1} -a^n z^{-n} = - \sum_{m=1}^{\infty} a^{-m} z^m = - \sum_{m=1}^{\infty} (a^{-1} z)^m$$

Similar to Example 5.1,  $X(z)$  converges if  $|a^{-1}z| < 1$  or  $|z| < |a|$ , which aligns with the ROC for  $X_-(z)$  in (5.17). This gives

$$X(z) = - \sum_{m=1}^{\infty} (a^{-1} z)^m = - \frac{a^{-1} z (1 - (a^{-1} z)^{\infty})}{1 - a^{-1} z} = - \frac{a^{-1} z}{1 - a^{-1} z} = \frac{z}{z - a}$$

Together with ROC, the  $z$  transform of  $x[n] = -a^n u[-n - 1]$  is:

$$X(z) = \frac{z}{z - a}, \quad |z| < |a|$$



Using (5.8), we substitute  $z = e^{j\omega}$  to obtain

$$X(e^{j\omega}) = \frac{e^{j\omega}}{e^{j\omega} - a}, \quad |e^{j\omega}| = 1 < |a|$$

As a result, the existence condition for DTFT of  $x[n]$  is  $|a| > 1$ .

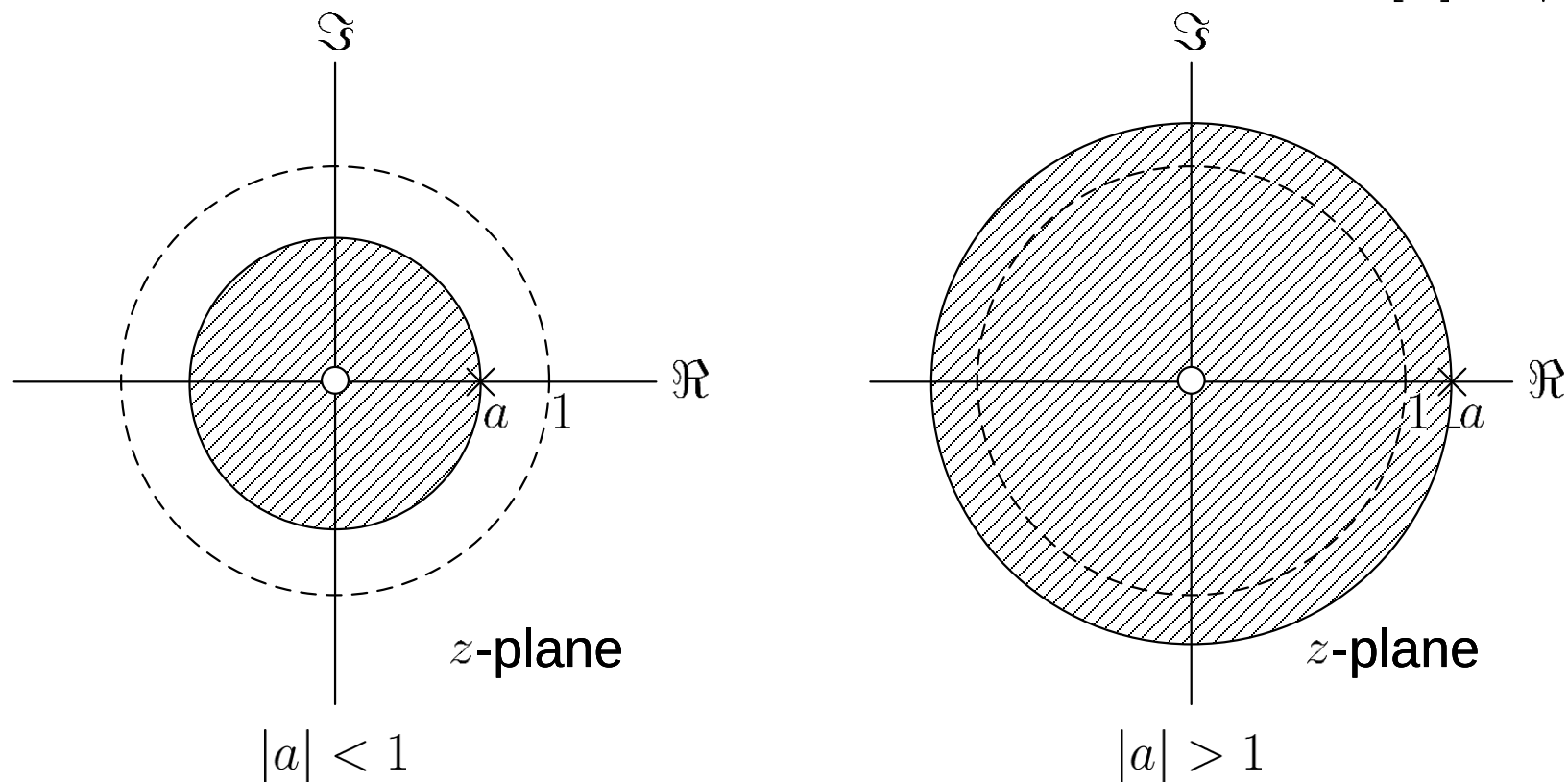


Fig.5.4: ROCs for  $|a| < 1$  and  $|a| > 1$  when  $x[n] = -a^n u[-n - 1]$

### Example 5.3

Determine the  $z$  transform of  $x[n] = a^n u[n] + b^n u[-n - 1]$  where  $|a| < |b|$ .

Employing the results in Examples 5.1 and 5.2, we have

$$\begin{aligned} X(z) &= \frac{1}{1 - az^{-1}} + \left( -\frac{1}{1 - bz^{-1}} \right), \quad |z| > |a| \quad \text{and} \quad |z| < |b| \\ &= \frac{(a - b)z^{-1}}{(1 - az^{-1})(1 - bz^{-1})} \\ &= \frac{(a - b)z}{(z - a)(z - b)}, \quad |a| < |z| < |b| \end{aligned}$$

Note that its ROC agrees with Fig.5.2.

**What are the pole(s) and zero(s) of  $X(z)$ ?**

### Example 5.4

Determine the  $z$  transform of  $x[n] = \delta[n + 1]$ .

Using (5.1) and (3.2), we have

$$X(z) = \sum_{n=-\infty}^{\infty} \delta[n + 1] z^{-n} = z$$

### Example 5.5

Determine the  $z$  transform of  $x[n]$  which has the form of:

$$x[n] = \begin{cases} a^n, & 0 \leq n \leq N - 1 \\ 0, & \text{otherwise} \end{cases}$$

Using (5.1), we have

$$X(z) = \sum_{n=0}^{N-1} (az^{-1})^n = \frac{1 - (az^{-1})^N}{1 - az^{-1}} = \frac{1}{z^{N-1}} \frac{z^N - a^N}{z - a}$$

**What are the ROCs in Examples 5.4 and 5.5?**

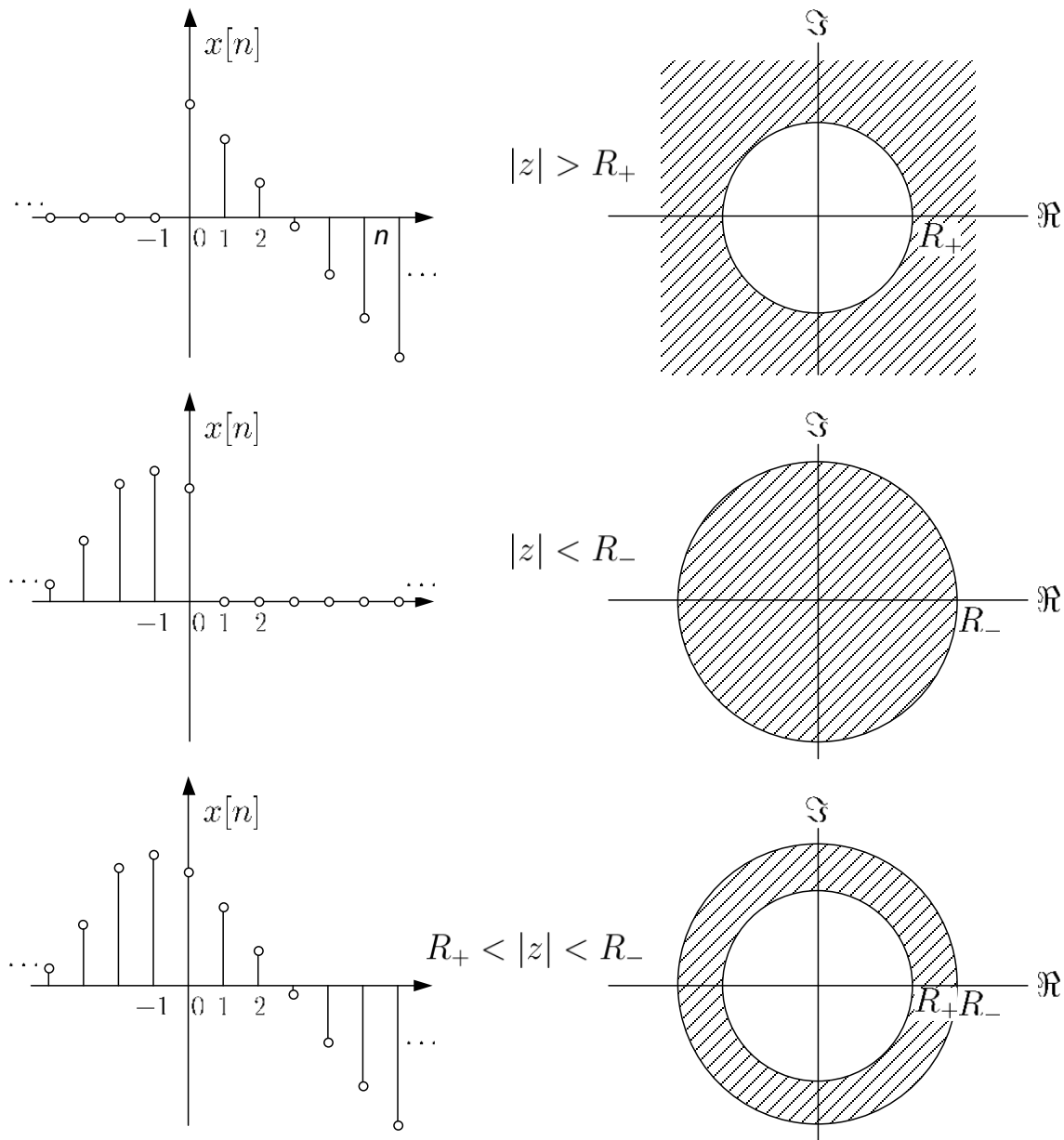


Figure 5.5: ROC for Infinite-duration sequences

## Z Transform of Finite-Duration Signals

For finite-duration signals the ZT is easy to find and the ROC is also easy to find because there are not infinitely many terms in the summation!

**Example:**  $x[n] = \left\{ \underset{\uparrow}{1}, 2, 5, 7, 0, 1 \right\}$  ↑ : position for  $n=0$

Finite Duration  
& Causal

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$$= \cdots + 0z^1 + 1z^0 + 2z^{-1} + 5z^{-2} + 7z^{-3} + 0z^{-4} + 1z^{-5} + 0z^{-6} + \cdots$$

$$= 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}$$

ROC = entire z plane except  $z = 0$

At  $z = 0$  we have  
terms that have  $1/0$

**Example:**  $x[n] = \left\{ 1, 2, \underset{\uparrow}{5}, 7, 0, 1 \right\}$

Finite Duration &  
"Bi-Directional"

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$$= 1z^2 + 2z^1 + 5 + 7z^{-1} + z^{-3}$$

ROC = entire z plane  
except  $z = 0$  and  $z = \infty$

**Example:**  $x[n] = \{1, 2, 5, 7, 0, 1\}$

Finite Duration  
& **Anti**-Causal

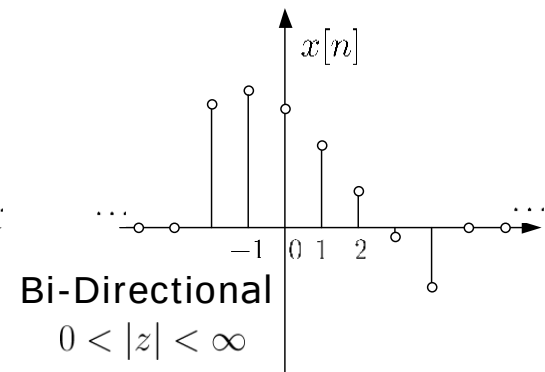
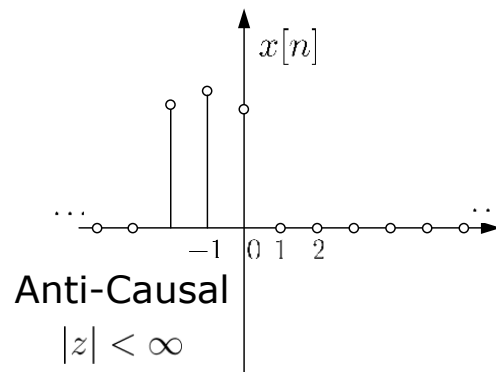
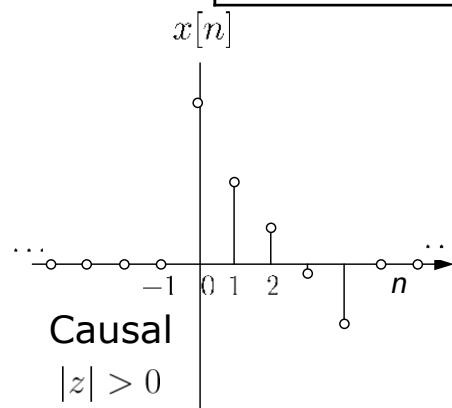
$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$$= 1z^5 + 2z^4 + 5z^3 + 7z^2 + 1$$

ROC = entire z plane  
except  $z = \infty$

## Summary of ROC for Finite-Duration Signals

| Type of <u>Finite-Duration</u> Signal | ROC                                   |
|---------------------------------------|---------------------------------------|
| Causal                                | All $z$ except $z = 0$                |
| Anti-Causal                           | All $z$ except $z = \infty$           |
| Bi-Directional                        | All $z$ except $z = \infty$ & $z = 0$ |



| Sequence           | Transform                                                       | ROC                                   |
|--------------------|-----------------------------------------------------------------|---------------------------------------|
| $\delta[n]$        | <b>1</b>                                                        | All $z$                               |
| $\delta[n - m]$    | $z^{-m}$                                                        | $ z  > 0, m > 0;  z  < \infty, m < 0$ |
| $a^n u[n]$         | $\frac{1}{1 - az^{-1}}$                                         | $ z  >  a $                           |
| $-a^n u[-n - 1]$   | $\frac{1}{1 - az^{-1}}$                                         | $ z  <  a $                           |
| $na^n u[n]$        | $\frac{az^{-1}}{(1 - az^{-1})^2}$                               | $ z  >  a $                           |
| $-na^n u[-n - 1]$  | $\frac{az^{-1}}{(1 - az^{-1})^2}$                               | $ z  <  a $                           |
| $a^n \cos(bn)u[n]$ | $\frac{1 - a \cos(b)z^{-1}}{1 - 2a \cos(b)z^{-1} + a^2 z^{-2}}$ | $ z  >  a $                           |
| $a^n \sin(bn)u[n]$ | $\frac{a \sin(b)z^{-1}}{1 - 2a \cos(b)z^{-1} + a^2 z^{-2}}$     | $ z  >  a $                           |

**Table 5.1:  $z$  transforms for common sequences**

Eight ROC properties are:

P1. There are four possible shapes for ROC, namely, the entire region except possibly  $z = 0$  and/or  $z = \infty$ , a ring, or inside or outside a circle in the  $z$ -plane centered at the origin (e.g., Figures 5.5 and 5.6)

P2. The DTFT of a sequence  $x[n]$  exists if and only if the ROC of the  $z$  transform of  $x[n]$  includes the unit circle (e.g., Examples 5.1 and 5.2)

P3: The ROC cannot contain any poles (e.g., Examples 5.1 to 5.5)

P4: When  $x[n]$  is a finite-duration sequence, the ROC is the entire  $z$ -plane except possibly  $z = 0$  and/or  $z = \infty$  (e.g., Examples 5.4 and 5.5)



P5: When  $x[n]$  is a right-sided sequence, the ROC is of the form  $|z| > |p_{\max}|$  where  $p_{\max}$  is the pole with the largest magnitude in  $X(z)$  (e.g., Example 5.1)

P6: When  $x[n]$  is a left-sided sequence, the ROC is of the form  $|z| < |p_{\min}|$  where  $p_{\min}$  is the pole with the smallest magnitude in  $X(z)$  (e.g., Example 5.2)

P7: When  $x[n]$  is a two-sided sequence, the ROC is of the form  $|p_a| < |z| < |p_b|$  where  $p_a$  and  $p_b$  are two poles with the successive magnitudes in  $X(z)$  such that  $|p_a| < |p_b|$  (e.g., Example 5.3)

P8: The ROC must be a connected region

### Example 5.6

A  $z$  transform  $X(z)$  contains three poles, namely,  $a$ ,  $b$  and  $c$  with  $|a| < |b| < |c|$ . Determine all possible ROCs.

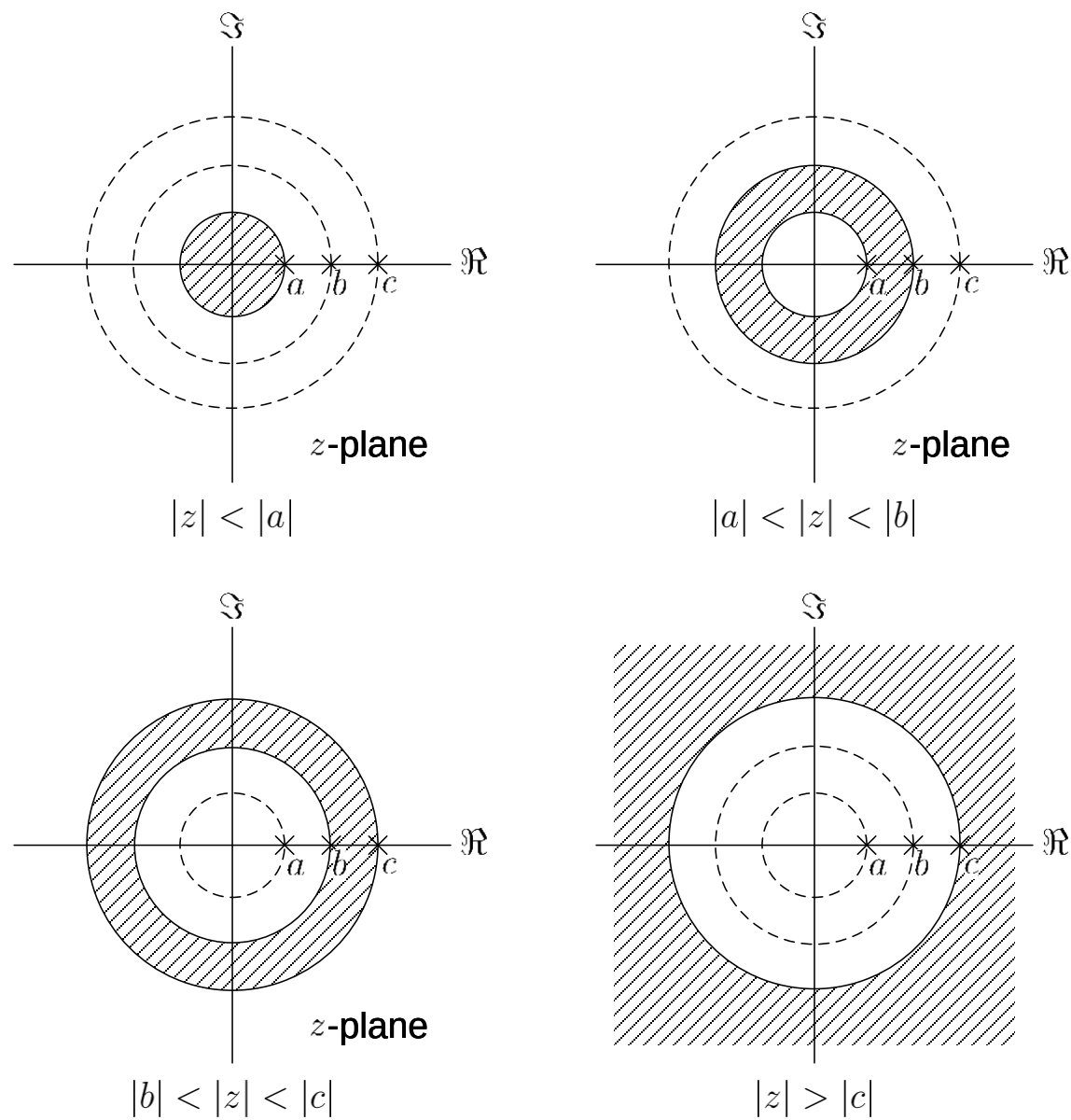


Fig.5.7: ROC possibilities for three poles

## Inverse z Transform

Inverse  $z$  transform corresponds to finding  $x[n]$  given  $X(z)$  and its ROC

$$x[n] = \frac{1}{2\pi j} \oint_{\Gamma} X[z] z^{n-1} dz \quad (5.20)$$

where  $\Gamma$  is a counterclockwise closed circular contour centered at the origin and with radius  $r$  such that  $\Gamma$  is in the ROC of  $X$ .

In practice, we do not usually compute the inverse  $z$  transform by directly using the formula from above.

There are 4 commonly used techniques to evaluate the inverse  $z$  transform. They are

1. Inspection
2. Partial Fraction Expansion
3. Power Series Expansion
4. Cauchy Integral Theorem

## Inspection

When we are familiar with certain transform pairs, we can do the inverse  $z$  transform by inspection

### Example 5.7

Determine the inverse  $z$  transform of  $X(z)$  which is expressed as:

$$X(z) = \frac{z}{2z - 1}, \quad |z| > 0.5$$

We first rewrite  $X(z)$  as:

$$X(z) = \frac{0.5}{1 - 0.5z^{-1}}$$

Making use of the following transform pair in Table 5.1:

$$a^n u[n] \leftrightarrow \frac{1}{1 - az^{-1}}, \quad |z| > |a|$$

and putting  $a = 0.5$ , we have:

$$\frac{0.5}{1 - 0.5z^{-1}} \leftrightarrow 0.5(0.5)^n u[n]$$

By inspection, the inverse  $z$  transform is:

$$x[n] = (0.5)^{n+1} u[n]$$

## Partial Fraction Expansion

Let us assume that  $X(z)$  is a ratio of polynomials in  $z^{-1}$  :

$$X(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} \quad (5.21)$$

For pole and zero determination, it is advantageous to multiply  $z^{M+N}$  to both numerator and denominator:

$$X(z) = \frac{z^N \sum_{k=0}^M b_k z^{M-k}}{z^M \sum_{k=0}^N a_k z^{N-k}} \quad (5.22)$$

- When  $M > N$ , there are  $(M - N)$  pole(s) at  $z = 0$
- When  $M < N$ , there are  $(N - M)$  zero(s) at  $z = 0$

To obtain the partial fraction expansion of  $X(z)$  in (5.21), it is most convenient to express  $X(z)$  in the form

$$X(z) = \frac{b_0 \prod_{k=1}^M (1 - d_k z^{-1})}{a_0 \prod_{k=1}^N (1 - c_k z^{-1})}$$

where the  $d_k$ 's are the nonzero zeros of  $X(z)$  and the  $c_k$ 's are the nonzero poles of  $X(z)$ .

There are 4 cases to be considered:

**Case 1:**  $M < N$  and all poles are of **first order**

For first-order poles, all  $\{c_k\}$  are distinct.  $X(z)$  is:

$$X(z) = \sum_{k=1}^N \frac{A_k}{1 - c_k z^{-1}} \quad (5.23)$$

For each first-order term of  $A_k / (1 - c_k z^{-1})$ , its inverse  $z$  transform can be easily obtained by inspection

Multiplying both sides by  $(1 - c_k z^{-1})$  and evaluating for  $z = c_k$

$$A_k = (1 - c_k z^{-1}) X(z) \Big|_{z=c_k} \quad (5.24)$$

An illustration for computing  $A_1$  with  $N = 2 > M$  is:

$$\begin{aligned} X(z) &= \frac{A_1}{1 - c_1 z^{-1}} + \frac{A_2}{1 - c_2 z^{-1}} \\ \Rightarrow (1 - c_1 z^{-1}) X(z) &= A_1 + \frac{A_2 (1 - c_1 z^{-1})}{1 - c_2 z^{-1}} \end{aligned} \quad (5.25)$$

Substituting  $z = c_1$ , we get  $A_1$

In summary, three steps are:

- Find poles
- Find  $\{A_k\}$
- Perform inverse  $z$  transform for the fractions by inspection



### Example 5.8

Find the pole and zero locations of  $H(z)$ :

$$H(z) = -\frac{1 + 0.1z^{-1}}{1 - 2.05z^{-1} + z^{-2}}$$

Then determine the inverse  $z$  transform of  $H(z)$ .

We first multiply  $z^2$  to both numerator and denominator polynomials to obtain:

$$H(z) = -\frac{z(z + 0.1)}{z^2 - 2.05z + 1}$$

Apparently, there are two zeros at  $z = 0$  and  $z = -0.1$ . On the other hand, by solving the quadratic equation at the denominator polynomial, the poles are determined as  $z = 0.8$  and  $z = 1.25$ .

According to (5.23), we have:

$$H(z) = \frac{A_1}{1 - 0.8z^{-1}} + \frac{A_2}{1 - 1.25z^{-1}}$$

Employing (5.24),  $A_1$  is calculated as:

$$A_1 = (1 - 0.8z^{-1}) H(z) \Big|_{z=0.8} = - \frac{1 + 0.1z^{-1}}{1 - 1.25z^{-1}} \Big|_{z=0.8} = 2$$

Similarly,  $A_2$  is found to be  $-3$ . As a result, the partial fraction expansion for  $H(z)$  is

$$H(z) = \frac{2}{1 - 0.8z^{-1}} - \frac{3}{1 - 1.25z^{-1}}$$

As the ROC is not specified, we investigate all possible scenarios, namely,  $|z| > 1.25$ ,  $0.8 < |z| < 1.25$ , and  $|z| < 0.8$ .

For  $|z| > 1.25$ , we notice that

$$(0.8)^n u[n] \leftrightarrow \frac{1}{1 - 0.8z^{-1}}, \quad |z| > 0.8$$

and

$$(1.25)^n u[n] \leftrightarrow \frac{1}{1 - 1.25z^{-1}}, \quad |z| > 1.25$$

where both ROCs agree with  $|z| > 1.25$ . Combining the results, the inverse  $z$  transform  $h[n]$  is:

$$h[n] = (2(0.8)^n - 3(1.25)^n) u[n]$$

which is a right-sided sequence and aligns with P5.

For  $0.8 < |z| < 1.25$ , we make use of

$$(0.8)^n u[n] \leftrightarrow \frac{1}{1 - 0.8z^{-1}}, \quad |z| > 0.8$$

and

$$-(1.25)^n u[-n-1] \leftrightarrow \frac{1}{1-1.25z^{-1}}, \quad |z| < 1.25$$

where both ROCs agree with  $0.8 < |z| < 1.25$ . This implies:

$$h[n] = 2(0.8)^n u[n] + 3(1.25)^n u[-n-1]$$

which is a two-sided sequence and aligns with P7.

Finally, for  $|z| < 0.8$ :

$$-(0.8)^n u[-n-1] \leftrightarrow \frac{1}{1-0.8z^{-1}}, \quad |z| < 0.8$$

and

$$-(1.25)^n u[-n-1] \leftrightarrow \frac{1}{1-1.25z^{-1}}, \quad |z| < 1.25$$

where both ROCs agree with  $|z| < 0.8$ . As a result, we have:

$$h[n] = (-2(0.8)^n + 3(1.25)^n) u[-n-1]$$

which is a left-sided sequence and aligns with P6.

Suppose  $h[n]$  is the impulse response of a discrete-time LTI system. Recall (3.11) and (3.12):

$$h[n] = 0, \quad n < 0$$

and

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

The three possible impulse responses:

- $h[n] = (2(0.8)^n - (1.25)^n) u[n]$  is the impulse response of a **causal** but **unstable** system
- $h[n] = 2(0.8)^n u[n] + (1.25)^n u[-n - 1]$  corresponds to a **noncausal** but **stable** system
- $h[n] = (-2(0.8)^n + (1.25)^n) u[-n - 1]$  is **noncausal** and **unstable**

**Which of the  $h[n]$  has / have DTFT?**

**Case 2:**  $M \geq N$  and all poles are of first order

In this case,  $X(z)$  can be expressed as:

$$X(z) = \sum_{l=0}^{M-N} B_l z^{-l} + \sum_{k=1}^N \frac{A_k}{1 - c_k z^{-1}} \quad (5.26)$$

- $B_l$  are obtained by **long division** of the numerator by the denominator, with the division process terminating when the remainder is of lower degree than the denominator
- $A_k$  can be obtained using (5.24).

### Example 5.9

Determine  $x[n]$  which has  $z$  transform of the form:

$$X(z) = \frac{4 - 2z^{-1} + z^{-2}}{1 - 1.5z^{-1} + 0.5z^{-2}}, \quad |z| > 1$$

The poles are easily determined as  $z = 0.5$  and  $z = 1$

According to (5.26) with  $M = N = 2$ :

$$X(z) = B_0 + \frac{A_1}{1 - 0.5z^{-1}} + \frac{A_2}{1 - z^{-1}}$$

The value of  $B_0$  is found by dividing the numerator polynomial by the denominator polynomial as follows:

$$\begin{array}{r} 0.5z^{-2} - 1.5z^{-1} + 1 \overline{) 2} \\ \underline{z^{-2} - 2z^{-1} + 4} \phantom{0} \\ - (z^{-2} - 3z^{-1} + 2) \\ \hline z^{-1} + 2 \end{array}$$

That is,  $B_0 = 2$ . Thus  $X(z)$  is expressed as

$$X(z) = 2 + \frac{2 + z^{-1}}{(1 - 0.5z^{-1})(1 - z^{-1})} = 2 + \frac{A_1}{1 - 0.5z^{-1}} + \frac{A_2}{1 - z^{-1}}$$

According to (5.24),  $A_1$  and  $A_2$  are calculated as

$$A_1 = \left. \frac{4 - 2z^{-1} + z^{-2}}{1 - z^{-1}} \right|_{z=0.5} = -4$$

and

$$A_2 = \left. \frac{4 - 2z^{-1} + z^{-2}}{1 - 0.5z^{-1}} \right|_{z=1} = 6$$

With  $|z| > 1$ :

$$\delta[n] \leftrightarrow 1$$

$$(0.5)^n u[n] \leftrightarrow \frac{1}{1 - 0.5z^{-1}}, \quad |z| > 0.5$$

and

$$u[n] \leftrightarrow \frac{1}{1 - z^{-1}}, \quad |z| > 1$$

the inverse  $z$  transform  $x[n]$  is:

$$x[n] = 2\delta[n] - 4(0.5)^n u[n] + 6u[n]$$



**Case 3:**  $M < N$  with multiple-order pole(s)

If  $X(z)$  has a  $s$ -order pole at  $z = c_i$  with  $s \geq 2$ , this means that there are  $s$  repeated poles with the same value of  $c_i$ .  $X(z)$  is:

$$X(z) = \sum_{k=1, k \neq i}^N \frac{A_k}{1 - c_k z^{-1}} + \sum_{m=1}^s \frac{C_m}{(1 - c_i z^{-1})^m} \quad (5.27)$$

- When there are two or more multiple-order poles, we include a component like the second term for each corresponding pole
- $A_k$  can be computed according to (5.24)
- $C_m$  can be calculated from:

$$C_m = \frac{1}{(s - m)!(-c_i)^{s-m}} \cdot \frac{d^{s-m}}{dw^{s-m}} \left[ (1 - c_i w)^s X(w^{-1}) \right] \bigg|_{w=c_i^{-1}} \quad (5.28)$$

### Example 5.10

Determine the partial fraction expansion for  $X(z)$ :

$$X(z) = \frac{4}{(1 + z^{-1})(1 - z^{-1})^2}$$

It is clear that  $X(z)$  corresponds to Case 3 with  $N = 3 > M$  and one second-order pole at  $z = 1$ . Hence  $X(z)$  is:

$$X(z) = \frac{A_1}{1 + z^{-1}} + \frac{C_1}{1 - z^{-1}} + \frac{C_2}{(1 - z^{-1})^2}$$

Employing (5.24),  $A_1$  is:

$$A_1 = \left. \frac{4}{(1 - z^{-1})^2} \right|_{z=-1} = 1$$

Applying (5.28),  $C_1$  is:

$$\begin{aligned} C_1 &= \frac{1}{(2-1)!(-1)^{2-1}} \cdot \frac{d}{dw} \left[ (1-1 \cdot w)^2 \frac{4}{(1+w)(1-w)^2} \right] \Big|_{w=1} \\ &= - \frac{d}{dw} \frac{4}{1+w} \Big|_{w=1} \\ &= \frac{4}{(1+w)^2} \Big|_{w=1} \\ &= 1 \end{aligned}$$

and

$$\begin{aligned} C_2 &= \frac{1}{(2-2)!(-1)^{2-2}} \cdot \left[ (1-1 \cdot w)^2 \frac{4}{(1+w)(1-w)^2} \right] \Big|_{w=1} \\ &= \frac{4}{1+w} \Big|_{w=1} \\ &= 2 \end{aligned}$$

Therefore, the partial fraction expansion for  $X(z)$  is

$$X(z) = \frac{1}{1 + z^{-1}} + \frac{1}{1 - z^{-1}} + \frac{2}{(1 - z^{-1})^2}$$

**Case 4:**  $M \geq N$  with multiple-order pole(s)

This is the most general case and the partial fraction expansion of  $X(z)$  is

$$X(z) = \sum_{l=0}^{M-N} B_l z^{-l} + \sum_{k=1, k \neq i}^N \frac{A_k}{1 - c_k z^{-1}} + \sum_{m=1}^s \frac{C_m}{(1 - c_i z^{-1})^m} \quad (5.29)$$

assuming that there is only one multiple-order pole of order  $s \geq 2$  at  $z = c_i$ . It is easily extended to the scenarios when there are two or more multiple-order poles as in Case 3. The  $A_k$ ,  $B_l$  and  $C_m$  can be calculated as in Cases 1, 2 and 3

## Power Series Expansion

When  $X(z)$  is expanded as power series according to (5.1):

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} = \cdots + x[-1]z^1 + x[0] + x[1]z^{-1} + x[2]z^{-2} + \cdots \quad (5.30)$$

any particular value of  $x[n]$  can be determined by finding the coefficient of the appropriate power of  $z^{-1}$

### Example 5.11

Determine  $x[n]$  which has  $z$  transform of the form:

$$X(z) = 2z^2 (1 - 0.5z^{-1}) (1 + z^{-1}) (1 - z^{-1}), \quad 0 < |z| < \infty$$

Expanding  $X(z)$  yields

$$X(z) = 2z^2 - z - 2 + z^{-1}$$

From (5.30),  $x[n]$  is deduced as:

$$x[n] = 2\delta[n+2] - \delta[n+1] - 2\delta[n] + \delta[n-1]$$

### Example 5.12

Determine  $x[n]$  whose  $z$  transform is given as:

$$X(z) = \log(1 + az^{-1}), \quad |z| > |a|$$

With the use of the power series expansion for  $\log(1 + \lambda)$ :

$$\log(1 + \lambda) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} \lambda^n}{n}, \quad |\lambda| < 1$$

$X(z)$  with  $|az^{-1}| < 1$  can be expressed as

$$\log(1 + az^{-1}) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} a^n z^{-n}}{n}$$

From (5.30),  $x[n]$  is deduced as:

$$x[n] = \frac{(-1)^{n+1} a^n}{n} u[n - 1]$$

### Example 5.13

Determine  $x[n]$  whose  $z$  transform has the form of:

$$X(z) = \frac{1}{1 - az^{-1}}, \quad |z| > |a|$$

With the use of

$$\frac{1}{1 - \lambda} = 1 + \lambda + \lambda^2 + \dots, \quad |\lambda| < 1$$

Carrying out long division in  $X(z)$  with  $|az^{-1}| < 1$ :

$$X(z) = 1 + az^{-1} + (az^{-1})^2 + \dots$$

From (5.30),  $x[n]$  is deduced as:

$$x[n] = a^n u[n]$$

which agrees with Example 5.1 and Table 5.1

### Example 5.14

Determine  $x[n]$  whose  $z$  transform has the form of:

$$X(z) = \frac{1}{1 - az^{-1}}, \quad |z| < |a|$$

We first express  $X(z)$  as:

$$X(z) = \frac{-a^{-1}z}{-a^{-1}z} \cdot \frac{1}{1 - az^{-1}} = \frac{-a^{-1}z}{1 - a^{-1}z}$$

Carrying out long division in  $X(z)$  with  $|a^{-1}z| < 1$ :

$$X(z) = -a^{-1}z \left( 1 + a^{-1}z + (a^{-1}z)^2 + \cdots \right)$$

From (5.30),  $x[n]$  is deduced as:

$$x[n] = -a^n u[-n - 1]$$

which agrees with Example 5.2 and Table 5.1



## Properties of z Transform

### 1. Linearity

Let  $(x_1[n], X_1(z))$  and  $(x_2[n], X_2(z))$  be two  $z$  transform pairs with ROCs  $\mathcal{R}_{x_1}$  and  $\mathcal{R}_{x_2}$ , respectively, we have

$$ax_1[n] + bx_2[n] \leftrightarrow aX_1(z) + bX_2(z) \quad (5.31)$$

Its ROC is denoted by  $\mathcal{R}$ , which **includes**  $\mathcal{R}_{x_1} \cap \mathcal{R}_{x_2}$  where  $\cap$  is the intersection operator. That is,  $\mathcal{R}$  **contains at least** the intersection of  $\mathcal{R}_{x_1}$  and  $\mathcal{R}_{x_2}$ .

### Example 5.15

Determine the  $z$  transform of  $y[n]$  which is expressed as:

$$y[n] = x_1[n] + x_2[n]$$

where  $x_1[n] = (0.2)^n u[n]$  and  $x_2[n] = (-0.3)^n u[n]$ . By inspection,

the  $z$  transforms of  $x_1[n]$  and  $x_2[n]$  are:

$$x_1[n] = (0.2)^n u[n] \leftrightarrow \frac{1}{1 - 0.2z^{-1}}, \quad |z| > 0.2$$

and

$$x_2[n] = (-0.3)^n u[n] \leftrightarrow \frac{1}{1 + 0.3z^{-1}}, \quad |z| > 0.3$$

According to the linearity property, the  $z$  transform of  $y[n]$  is

$$Y(z) = \frac{1}{1 - 0.2z^{-1}} + \frac{1}{1 + 0.3z^{-1}}, \quad |z| > 0.3$$

**Why the ROC is  $|z| > 0.3$  instead of  $|z| > 0.2$ ?**

## 2. Time Shifting

A time-shift of  $n_0$  in  $x[n]$  causes a multiplication of  $z^{-n_0}$  in  $X(z)$

$$x[n - n_0] \leftrightarrow z^{-n_0} X(z) \quad (5.32)$$

The ROC for  $x[n - n_0]$  is basically identical to that of  $X(z)$  except for the possible addition or deletion of  $z = 0$  or  $z = \infty$

### Example 5.16

Find the  $z$  transform of  $x[n]$  which has the form of:

$$x[n] = a^{n-1}u[n-1]$$

Employing the time-shifting property with  $n_0 = 1$  and:

$$a^n u[n] \leftrightarrow \frac{1}{1 - az^{-1}}, \quad |z| > |a|$$

we easily obtain

$$a^{n-1}u[n-1] \leftrightarrow z^{-1} \cdot \frac{1}{1 - az^{-1}} = \frac{z^{-1}}{1 - az^{-1}}, \quad |z| > |a|$$

Note that using (5.1) with  $|z| > |a|$  also produces the same result but this approach is less efficient:

$$X(z) = \sum_{n=1}^{\infty} a^{n-1}z^{-n} = a^{-1} \sum_{n=1}^{\infty} (az^{-1})^n = a^{-1} \frac{az^{-1} [1 - (az^{-1})^{\infty}]}{1 - az^{-1}} = \frac{z^{-1}}{1 - az^{-1}}$$

### 3. Multiplication by an Exponential Sequence (Modulation)

If we multiply  $x[n]$  by  $z_0^n$  in the time domain, the variable  $z$  will be changed to  $z/z_0$  in the  $z$  transform domain. That is:

$$z_0^n x[n] \leftrightarrow X(z/z_0) \quad (5.33)$$

If the ROC for  $x[n]$  is  $R_+ < |z| < R_-$ , the ROC for  $z_0^n x[n]$  is  $|z_0|R_+ < |z| < |z_0|R_-$

#### Example 5.17

With the use of the following  $z$  transform pair:

$$u[n] \leftrightarrow \frac{1}{1 - z^{-1}}, \quad |z| > 1$$

Find the  $z$  transform of  $x[n]$  which has the form of:

$$x[n] = a^n \cos(bn)u[n]$$

Noting that  $\cos(bn) = (e^{jbn} + e^{-jbn})/2$ ,  $x[n]$  can be written as:

$$x[n] = \frac{1}{2} (ae^{jb})^n u[n] + \frac{1}{2} (ae^{-jb})^n u[n]$$

By means of the modulation property of (5.33) with the substitution of  $z_0 = ae^{jb}$  and  $z_0 = ae^{-jb}$ , we obtain:

$$\frac{1}{2} (ae^{jb})^n u[n] \leftrightarrow \frac{1}{2} \frac{1}{1 - (z/(ae^{jb}))^{-1}} = \frac{1}{2} \frac{1}{1 - ae^{jb}z^{-1}}, \quad |z| > |a|$$

and

$$\frac{1}{2} (ae^{-jb})^n u[n] \leftrightarrow \frac{1}{2} \frac{1}{1 - (z/(ae^{-jb}))^{-1}} = \frac{1}{2} \frac{1}{1 - ae^{-jb}z^{-1}}, \quad |z| > |a|$$

By means of the linearity property, it follows that

$$X(z) = \frac{1}{2} \frac{1}{1 - ae^{jb}z^{-1}} + \frac{1}{2} \frac{1}{1 - ae^{-jb}z^{-1}} = \frac{1 - a \cos(b)z^{-1}}{1 - 2a \cos(b)z^{-1} + a^2z^{-2}}, \quad |z| > |a|$$

which agrees with Table 5.1.

## 4. Differentiation

Differentiating  $X(z)$  with respect to  $z$  corresponds to multiplying  $x[n]$  by  $n$  in the time domain:

$$nx[n] \leftrightarrow -z \frac{dX(z)}{dz} \quad (5.34)$$

The ROC for  $nx[n]$  is basically identical to that of  $X(z)$  except for the possible addition or deletion of  $z = 0$  or  $z = \infty$

### Example 5.18

Determine the  $z$  transform of  $x[n] = na^n u[n]$ .

Since

$$a^n u[n] \leftrightarrow \frac{1}{1 - az^{-1}}, \quad |z| > |a|$$

and

$$\frac{d}{dz} \left( \frac{1}{1 - az^{-1}} \right) = \frac{d(1 - az^{-1})^{-1}}{d(1 - az^{-1})} \cdot \frac{d(1 - az^{-1})}{dz} = -\frac{az^{-2}}{(1 - az^{-1})^2}$$

By means of the differentiation property, we have

$$na^n u[n] \leftrightarrow -z \cdot -\frac{az^{-2}}{(1 - az^{-1})^2} = \frac{az^{-1}}{(1 - az^{-1})^2}, \quad |z| > |a|$$

which agrees with Table 5.1.

## 5. Conjugation

The  $z$  transform pair for  $x^*[n]$  is:

$$x^*[n] \leftrightarrow X^*(z^*) \quad (5.35)$$

The ROC for  $x^*[n]$  is identical to that of  $x[n]$

## 6. Time Reversal

The  $z$  transform pair for  $x[-n]$  is:

$$x[-n] \leftrightarrow X(z^{-1}) \quad (5.36)$$

If the ROC for  $x[n]$  is  $R_+ < |z| < R_-$ , the ROC for  $x[-n]$  is  $1/R_- < |z| < 1/R_+$

### Example 5.19

Determine the  $z$  transform of  $x[n] = -na^{-n}u[-n]$

Using Example 5.18:

$$na^n u[n] \leftrightarrow \frac{az^{-1}}{(1 - az^{-1})^2}, \quad |z| > |a|$$

and from the time reversal property:

$$X(z) = \frac{az}{(1 - az)^2} = \frac{a^{-1}z^{-1}}{(1 - a^{-1}z^{-1})^2}, \quad |z| < |a^{-1}|$$



## 7. Convolution

Let  $(x_1[n], X_1(z))$  and  $(x_2[n], X_2(z))$  be two  $z$  transform pairs with ROCs  $\mathcal{R}_{x_1}$  and  $\mathcal{R}_{x_2}$ , respectively. Then we have:

$$x_1[n] \otimes x_2[n] \leftrightarrow X_1(z)X_2(z) \quad (5.37)$$

and its ROC includes  $\mathcal{R}_{x_1} \cap \mathcal{R}_{x_2}$ .

The proof is given as follows.

Let

$$y[n] = x_1[n] \otimes x_2[n] = \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k] \quad (5.38)$$

With the use of the time shifting property,  $Y(z)$  is:

$$\begin{aligned}
Y(z) &= \sum_{n=-\infty}^{\infty} \left[ \sum_{k=-\infty}^{\infty} x_1[k]x_2[n-k] \right] z^{-n} \\
&= \sum_{k=-\infty}^{\infty} x_1[k] \left[ \sum_{n=-\infty}^{\infty} x_2[n-k]z^{-n} \right] \\
&= \sum_{k=-\infty}^{\infty} x_1[k]X_2(z)z^{-k} \\
&= X_1(z)X_2(z)
\end{aligned} \tag{5.39}$$

## Transfer Function of Linear Time-Invariant System

A LTI system can be characterized by the **transfer function**, which is a  $z$  transform expression

Starting with:

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k] \quad (5.40)$$

Applying  $z$  transform on (5.40) with the use of the linearity and time shifting properties, we have

$$Y(z) \sum_{k=0}^N a_k z^{-k} = X(z) \sum_{k=0}^M b_k z^{-k} \quad (5.41)$$

The transfer function, denoted by  $H(z)$ , is defined as:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} \quad (5.42)$$

The system impulse response  $h[n]$  is given by the inverse  $z$  transform of  $H(z)$  with an appropriate ROC, that is,  $h[n] \leftrightarrow H(z)$ , such that  $y[n] = x[n] \otimes h[n]$ . This suggests that we can first take the  $z$  transforms for  $x[n]$  and  $h[n]$ , then multiply  $X(z)$  by  $H(z)$ , and finally perform the inverse  $z$  transform of  $X(z)H(z)$ .

### Example 5.20

Determine the transfer function for a LTI system whose input  $x[n]$  and output  $y[n]$  are related by:

$$y[n] = 0.1y[n-1] + x[n] + x[n-1]$$

Applying  $z$  transform on the difference equation with the use of the linearity and time shifting properties,  $H(z)$  is:

$$Y(z) (1 - 0.1z^{-1}) = X(z) (1 + z^{-1}) \Rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{1 + z^{-1}}{1 - 0.1z^{-1}}$$

Note that there are two ROC possibilities, namely,  $|z| > 0.1$  and  $|z| < 0.1$  and we cannot uniquely determine  $h[n]$

### Example 5.21

Find the difference equation of a LTI system whose transfer function is given by

$$H(z) = \frac{(1 + z^{-1})(1 - 2z^{-1})}{(1 - 0.5z^{-1})(1 + 2z^{-1})}$$

Let  $H(z) = Y(z)/X(z)$ . Performing cross-multiplication and inverse  $z$  transform, we obtain:

$$\begin{aligned}(1 - 0.5z^{-1})(1 + 2z^{-1})Y(z) &= (1 + z^{-1})(1 - 2z^{-1})X(z) \\ \Rightarrow (1 + 1.5z^{-1} - z^{-2})Y(z) &= (1 - z^{-1} - 2z^{-2})X(z) \\ \Rightarrow y[n] + 1.5y[n-1] - y[n-2] &= x[n] - x[n-1] - 2x[n-2]\end{aligned}$$

Examples 5.20 and 5.21 imply the equivalence between the difference equation and transfer function

### Example 5.22

Compute the impulse response  $h[n]$  for a LTI system which is characterized by the following difference equation:

$$y[n] = x[n] - x[n - 1]$$

Applying  $z$  transform on the difference equation with the use of the linearity and time shifting properties,  $H(z)$  is:

$$Y(z) = X(z) (1 - z^{-1}) \Rightarrow H(z) = \frac{Y(z)}{X(z)} = 1 - z^{-1}$$

There is only one ROC possibility, namely,  $|z| > 0$ . Taking the inverse  $z$  transform on  $H(z)$ , we get:

$$h[n] = \delta[n] - \delta[n - 1]$$

which agrees with Example 3.10

### Example 5.23

Determine the output  $y[n]$  if the input is  $x[n] = u[n]$  and the LTI system impulse response is  $h[n] = \delta[n] + 0.5\delta[n - 1]$

The  $z$  transforms for  $x[n]$  and  $h[n]$  are

$$X(z) = \frac{1}{1 - z^{-1}}, \quad |z| > 1$$

and

$$H(z) = 1 + 0.5z^{-1} \quad |z| > 0$$

As a result, we have:

$$Y(z) = X(z)H(z) = \frac{1}{1 - z^{-1}} + 0.5\frac{z^{-1}}{1 - z^{-1}}, \quad |z| > 1$$

Taking the inverse  $z$  transform of  $Y(z)$  with the use of the time shifting property yields:

$$y[n] = u[n] + 0.5u[n - 1]$$